

Engineering Mechanics: Kinematics Solutions

Question One

(a) Rectilinear motion with constant acceleration

For rectilinear motion with constant acceleration $a = \text{constant}$, the following kinematic equations describe the motion:

$$v = v_0 + at \quad (\text{velocity as function of time}) \quad (1)$$

$$s = s_0 + v_0 t + \frac{1}{2}at^2 \quad (\text{position as function of time}) \quad (2)$$

$$v^2 = v_0^2 + 2a(s - s_0) \quad (\text{velocity as function of position}) \quad (3)$$

These equations allow analysis of:

- Time to reach a certain velocity
- Distance traveled in a given time
- Final velocity after a known displacement
- Stopping distance under constant deceleration

Graphical analysis may include:

- v - t graph: straight line with slope a
- s - t graph: parabola
- a - t graph: horizontal line

Area under v - t curve gives displacement; slope of v - t curve gives acceleration.

(b) Rocket motion under gravity

Given:

- Initial velocity at engine failure: $v_A = 50 \text{ m/s}$ (upward)
- Position at engine failure: $S_A = 40 \text{ m}$ above ground
- Acceleration after failure: $a = -g = -9.81 \text{ m/s}^2$
- Neglect air resistance

(i) Maximum height S_B reached by rocket

At maximum height, $v_B = 0$. Using $v^2 = v_0^2 + 2a(s - s_0)$:

$$0 = (50)^2 + 2(-9.81)(S_B - 40)$$

$$0 = 2500 - 19.62(S_B - 40)$$

$$19.62(S_B - 40) = 2500$$

$$S_B - 40 = \frac{2500}{19.62} \approx 127.42$$

$$S_B \approx 167.42 \text{ m}$$

(ii) Speed just before hitting the ground

From maximum height $S_B = 167.42$ m to ground $S = 0$:

Initial velocity at S_B is 0, downward displacement = 167.42 m, $a = g = 9.81 \text{ m/s}^2$:

$$v^2 = 0 + 2(9.81)(167.42)$$

$$v^2 \approx 3283.0$$

$$v \approx 57.30 \text{ m/s}$$

(iii) Total time of flight

Case 1: From when flight started (assuming launch from ground with initial velocity $v_0 = 50$ upward)

$$s(t) = 0 + 50t - \frac{1}{2}(9.81)t^2$$

At ground impact, $s = 0$:

$$0 = 50t - 4.905t^2$$

$$t(50 - 4.905t) = 0$$

$$t = 0 \quad \text{or} \quad t = \frac{50}{4.905} \approx 10.19 \text{ s}$$

So total flight time from launch ≈ 10.19 s.

Case 2: From when engine fails (at $S_A = 40$ m, $v_A = 50$ m/s upward)

Time from failure to max height:

$$v_B = v_A - gt_1 \implies 0 = 50 - 9.81t_1 \implies t_1 \approx 5.097 \text{ s}$$

Time from max height to ground:

$$S_B = \frac{1}{2}gt_2^2 \implies 167.42 = 0.5 \times 9.81 \times t_2^2$$

$$t_2^2 = \frac{167.42}{4.905} \approx 34.13 \implies t_2 \approx 5.842 \text{ s}$$

Total time from engine failure to impact:

$$t_{\text{total, after failure}} = t_1 + t_2 \approx 5.097 + 5.842 \approx 10.94 \text{ s}$$

Answers for Question One

$S_B \approx 167.42 \text{ m}, \quad v_{\text{impact}} \approx 57.30 \text{ m/s}, \quad t_{\text{total, launch}} \approx 10.19 \text{ s}, \quad t_{\text{total, after failure}} \approx 10.94 \text{ s}$
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Question Two

(a) Curvilinear motion analysis using rectangular components

In curvilinear motion, the position vector of a particle at time t is:

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$$

Velocity and acceleration:

$$\mathbf{v}(t) = \dot{x}\mathbf{i} + \dot{y}\mathbf{j} + \dot{z}\mathbf{k} = v_x\mathbf{i} + v_y\mathbf{j} + v_z\mathbf{k}$$

$$\mathbf{a}(t) = \ddot{x}\mathbf{i} + \ddot{y}\mathbf{j} + \ddot{z}\mathbf{k} = a_x\mathbf{i} + a_y\mathbf{j} + a_z\mathbf{k}$$

Magnitudes:

$$v = \sqrt{v_x^2 + v_y^2 + v_z^2}, \quad a = \sqrt{a_x^2 + a_y^2 + a_z^2}$$

Possible kinematic analyses:

1. Position from velocity: $\mathbf{r}(t) = \int \mathbf{v}(t) dt + \mathbf{r}_0$
2. Velocity from acceleration: $\mathbf{v}(t) = \int \mathbf{a}(t) dt + \mathbf{v}_0$
3. Independent scalar motion along each axis
4. Tangential/normal components from $\mathbf{v}$ and $\mathbf{a}$

(b) Given velocity vector

$$\mathbf{v}(t) = (16t^2)\mathbf{i} - (4t^3)\mathbf{j} + (5t + 2)\mathbf{k} \quad \text{m/s}$$

Particle starts at origin when $t = 0$: $x(0) = 0, y(0) = 0, z(0) = 0$.

(i) Velocity magnitude at $t = 4$ s

$$v_x(4) = 256, \quad v_y(4) = -256, \quad v_z(4) = 22$$

$$v(4) = \sqrt{256^2 + (-256)^2 + 22^2} = \sqrt{65536 + 65536 + 484} = \sqrt{131556} \approx 362.71 \text{ m/s}$$

(ii) Acceleration magnitude at $t = 4$ s

$$\mathbf{a}(t) = (32t)\mathbf{i} - (12t^2)\mathbf{j} + 5\mathbf{k}$$

$$a_x(4) = 128, \quad a_y(4) = -192, \quad a_z(4) = 5$$

$$a(4) = \sqrt{128^2 + (-192)^2 + 5^2} = \sqrt{16384 + 36864 + 25} = \sqrt{53273} \approx 230.83 \text{ m/s}^2$$

(iii) Position at $t = 4$ s

$$x(t) = \frac{16t^3}{3}, \quad x(4) = \frac{1024}{3} \approx 341.33 \text{ m}$$

$$y(t) = -t^4, \quad y(4) = -256 \text{ m}$$

$$z(t) = \frac{5t^2}{2} + 2t, \quad z(4) = 48 \text{ m}$$

$$\mathbf{r}(4) = (341.33, -256, 48) \text{ m}$$

Answers for Question Two

$v(4) \approx 362.71 \text{ m/s}, \quad a(4) \approx 230.83 \text{ m/s}^2, \quad \mathbf{r}(4) = (341.33, -256, 48) \text{ m}$
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Question Three

(a) Projectile motion from first principles

(i) Horizontal motion

$$a_x = 0 \Rightarrow v_x = (v_0)_x = \text{constant}$$
$$x(t) = x_0 + (v_0)_x \cdot t$$

(ii) Vertical motion

$$a_y = -g \quad (\text{downward})$$
$$v_y(t) = (v_0)_y - gt$$
$$y(t) = y_0 + (v_0)_y t - \frac{1}{2}gt^2$$

Maximum height when $v_y = 0 \Rightarrow t = \frac{(v_0)_y}{g}$:

$$y_{\max} = y_0 + \frac{(v_0)_y^2}{2g}$$

(b) Water pumped at 45° from height 2 m, lands after 3 s

Let u = initial speed. Then:

$$u_x = u \cos 45^\circ = \frac{u}{\sqrt{2}}, \quad u_y = u \sin 45^\circ = \frac{u}{\sqrt{2}}$$

(i) Exit speed

Vertical motion: $y(t) = y_0 + u_y t - \frac{1}{2}gt^2$, with $y_0 = 2$, $g = 9.81$, $t = 3$, $y(3) = 0$:

$$0 = 2 + \frac{3u}{\sqrt{2}} - \frac{1}{2}(9.81)(9)$$
$$0 = 2 + \frac{3u}{\sqrt{2}} - 44.145$$
$$\frac{3u}{\sqrt{2}} = 42.145$$
$$u = \frac{42.145 \times \sqrt{2}}{3} \approx 19.87 \text{ m/s}$$

(ii) Horizontal distance

$$u_x = \frac{u}{\sqrt{2}} \approx 14.05 \text{ m/s}$$
$$R = u_x \cdot t \approx 14.05 \times 3 \approx 42.15 \text{ m}$$

(iii) Maximum height

Time to max height: $t_m = \frac{u_y}{g} = \frac{14.05}{9.81} \approx 1.432 \text{ s}$ Height above launch:

$$h = u_y t_m - \frac{1}{2}gt_m^2 \approx 20.12 - 10.06 = 10.06 \text{ m}$$
$$H = 2 + h \approx 12.06 \text{ m}$$

(iv) Velocity components at ground impact

$$v_x(t = 3) = u_x \approx 14.05 \text{ m/s}$$

$$v_y(t = 3) = u_y - gt \approx 14.05 - 29.43 = -15.38 \text{ m/s}$$

$$\boxed{(14.05, -15.38) \text{ m/s}}$$

Answers for Question Three

(i) $u \approx 19.87 \text{ m/s}$

(ii) $R \approx 42.15 \text{ m}$

(iii) $H \approx 12.06 \text{ m}$

(iv) $(v_x, v_y) \approx (14.05, -15.38) \text{ m/s}$